

IMPORTS

```
import pandas as pd
import glob
```

PIVOT TABLE

```
weather_melt.pivot_table(index=['id','year','month','day'], columns='element', values='temp')
weather_melt.pivot_table(..., aggfunc='mean')
weather_tidy =
    weather_melt.pivot_table(...).reset_index()
df.pivot_table(index='id', columns='var', values='val').reset_index()
```

MELT BASIC

```
pd.melt(pew, id_vars='religion')
pd.melt(pew, id_vars='religion', var_name='income', value_name='count')
```

NORMALIZE

```
billboard_songs = billboard_long[['year','artist','track','time']].drop_duplicates()
billboard_songs['id'] = range(len(billboard_songs))
billboard_ratings =
    billboard_long.merge(billboard_songs, on=['year','artist','track','time'])
billboard_ratings = billboard_ratings[['id','date.entered','week','rating']]
```

MELT MULTI ID

```
pd.melt(billboard, id_vars=['year','artist','track','time','date.entered'], var_name='week', value_name='rating')
```

MULTIPLE FILES

```
glob.glob('data/*.csv')
files = glob.glob('data/*.csv')
pd.read_csv(f)
dfs = [pd.read_csv(f) for f in files]
pd.concat(dfs)
pd.concat([pd.read_csv(f) for f in files], ignore_index=True)
```

MELT WEATHER

```
weather_melt = pd.melt(weather, id_vars=['id','year','month','element'], var_name='day', value_name='temp')
```

RESHAPING PAIRS

```
wide -> pd.melt(...)
long -> df.pivot_table(...)
split -> s.str.split(...)
attach -> pd.concat(..., axis=1)
dedupe -> df.drop_duplicates()
link -> df.merge(...)
```

SPLIT STRINGS

```
variable_split = ebola_long['variable'].str.split('_')
ebola_long['status'] = variable_split.str.get(0)
ebola_long['country'] = variable_split.str.get(1)
variable_split = ebola_long['variable'].str.split('_', expand=True)
variable_split.columns = ['status','country']
pd.concat([ebola_long, variable_split], axis=1)
ebola_long['status'], ebola_long['country'] = zip(*ebola_long['variable'].str.split('_'))
```